Lab 2 - Configuring Basic Single-Area OSPFv2

Answer Sheet

Name: ___________Ma MaoHeng______________

Student ID: ____________22102017D_____________

Part 2: Configure and Verify OSPF Routing

Step 1: Verify OSPF neighbors and routing information.

b.

What command would you use to only see the OSPF routes in the routing table?

show ip route ospf

2

Part 4: Configure OSPF Passive Interfaces

g. Re-issue the **show ip route** and **show ip ospf neighbor** commands on R1 and R3, and look for a route to the 192.168.2.0/24 network.

What interface is R3 using to route to the 192.168.2.0/24 network? ___Serial0/0/0 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3? ___129__ 2

Does R2 show up as an OSPF neighbor on R1? ___yes_____ 2

Does R2 show up as an OSPF neighbor on R3? ___no_____ 2

What does this information tell you?

R2's S0/0/1 interface is passive (no Hellos sent), so no adjacency forms with R3. R3 must reach R2's LAN via R1 (R1 still has full adjacency with R2). 2

h. Change interface S0/0/1 on R2 to allow it to advertise OSPF routes. Record the commands used below.

R2(config)# router ospf 1

R2(config-router)# no passive-interface s0/0/1 2

---

i. Re-issue the **show ip route** command on R3.

What interface is R3 using to route to the 192.168.2.0/24 network? _____Serial0/0/1 2

What is the accumulated cost metric for the 192.168.2.0/24 network on R3 now and how is this calculated?

_Cost = 65. R3 S0/0/1 cost (64) + R2 G0/0 cost (1) = 65. This is lower than via R1. 2

Is R2 listed as an OSPF neighbor to R3? ____yes____ 2

Part 5: Change OSPF Metrics

Step 1: Change the reference bandwidth on the routers.

a. To reset the reference-bandwidth back to its default value, issue the **auto-cost reference-bandwidth 100** command on all three routers.

```
R1(config)# router ospf 1
R1(config-router)# auto-cost reference-bandwidth 100
% OSPF: Reference bandwidth is changed.
        Please ensure reference bandwidth is consistent across all routers.
```

Why would you want to change the OSPF default reference-bandwidth?

To correctly differentiate link costs for modern networks with speeds >100 Mbps (Fast Ethernet, Gigabit Ethernet, etc.). With the default 100 Mbps reference, all links ≥100 Mbps receive a cost of 1; increasing the reference bandwidth gives higher-speed links appropriately lower costs. 2

Step 2: Change the bandwidth for an interface.

g.

Explain how the costs to the 192.168.3.0/24 and 192.168.23.0/30 networks from R1 were calculated.

192.168.3.0/24: R1 S0/0/1 cost (781) + R3 G0/0 cost (1) = 782.

192.168.23.0/30: R1 S0/0/1 cost (781) + R3 S0/0/0 cost (64) = 845, 2

or R1 S0/0/0 cost (781) + R2 S0/0/1 cost (64) = 845. Both paths have equal cost.

i. Issue the **bandwidth 128** command on all remaining serial interfaces in the topology.

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why?

What is the new accumulated cost to the 192.168.23.0/24 network on R1? Why? 0

The new cost is 845. Cost = R1 S0/0/1 (781) + R3 S0/0/0 (64) = 845 (or via R2 with same cost). After applying bandwidth 128 on all serial interfaces, costs remain the same because all serial links now have consistent bandwidth settings.

Step 3: Change the route cost.

c.

Explain why the route to the 192.168.3.0/24 network on R1 is now going through R2?

The ip ospf cost 1565 was applied to R1 S0/0/1, making the direct path to R3 cost = 1565 + 1 = 1566. The path through R2 via S0/0/0 costs 781 + 781 + 1 = 1563, which is lower. OSPF always selects the path with the lowest accumulated cost.

4

Reflection

1. Why is it important to control the router ID assignment when using the OSPF protocol?

   The router ID serves as a unique identifier for each router inside the OSPF domain. It plays a key role in the DR/BDR election process on multi-access networks and is also used as the originator identifier in LSAs. If left to automatic selection, the router ID defaults to the highest IP address among active interfaces (usually a physical one). This can lead to instability—if that interface fails or its IP changes, the router ID may change, potentially disrupting adjacencies, elections, or the LSDB. Manually setting a stable router ID (typically via a loopback interface or the explicit router-id command) ensures predictability and consistency across OSPF operations.

   4

2. Why is the DR/BDR election process not a concern in this lab?

   DR and BDR elections only take place on multi-access (broadcast) network types, such as Ethernet LANs, where multiple routers share the same segment. In this lab, the connections between routers use serial links, which are treated as point-to-point networks—OSPF skips the DR/BDR process entirely on these. Additionally, the few broadcast-type interfaces (like G0/0 LANs) each connect to only a single router, so no election actually occurs or has any meaningful impact.

   4

3. Why would you want to set an OSPF interface to passive?

   Marking an interface as passive prevents OSPF from sending Hello packets out of it, while still allowing the connected network to be advertised into OSPF. This is especially useful on LAN interfaces facing end users or devices that don't run OSPF—no neighbor relationships will form there. Benefits include reduced unnecessary protocol traffic (less overhead), cleaner network operation, and improved security by blocking the possibility of rogue or unauthorized devices attempting to form OSPF adjacencies.

   4

4. What aspect of OSPF makes the protocol hierarchical and permits the creation of very scalable networks?

   OSPF achieves hierarchy and excellent scalability through the use of areas. The protocol organizes the network into multiple areas, with Area 0 serving as the mandatory backbone that connects all other areas. This design limits the scope of LSA flooding and SPF (Dijkstra) calculations—changes in one area don't flood everywhere, routers maintain smaller link-state databases, and overall CPU/memory demands stay manageable even in very large deployments.

   4

5. What single OSPF router configuration command allows the assignment of OSPF area 0 to all interfaces in the range 10.0.0.0 to 10.255.255.255?

range 10.0.0.0–10.255.255.255 to OSPF Area 0?network 10.0.0.0 0.255.255.255 area 0 (The wildcard mask 0.255.255.255 matches any address where the first octet is 10 and the remaining three octets can be anything from 0–255.)

6. A router has three routes to the same network: one route from a RIP with a metric of 4, another from OSPF with a metric of 3053092, and another from EIGRP with a metric of 4039043. Which of the three routes will be used for the routing decision?

   The EIGRP route will be selected. Routing protocol path selection on Cisco routers prioritizes administrative distance (AD) over the protocol's own metric. Default AD values are: EIGRP = 90 (internal), OSPF = 110, RIP = 120. Since EIGRP has the lowest (most trusted) AD, its route wins regardless of how high its metric appears compared to the others.